Finite-time memory from two returns
A controlled reduction to ordinary charge propagation

Alan Day Chengheng*

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Abstract

For the common-coupling complex SYK charge-removal dilation, we reduce the actual finite-memory covariance to an auxiliary kernel with ordinary Hamiltonian propagation. A finite Volterra iteration and uniform Gaussian force moments give a two-return approximation with collective-field error $O(p^{-2})$ and absolute covariance error $O(p^{-1})$. Time exponentials are kept intact. The remaining task is the limit of the resulting pinched, common-coupling products.

1 State, operators and the result

Use standard CAR, $\hbar = 1$, string-time units, and $H = \sum_{I,J} J_{IJ} c_I^\dagger c_J$, where I, J are increasing p-subsets, $c_J = (c_J^\dagger)^\dagger$, $J = J^\dagger$ is Gaussian, and

$$\mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \qquad w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \quad \lambda_N = p^2/N. \tag{1}$$

Take even $N \to \infty$, $\lambda_N \in [\lambda_-, \lambda_+] \subset (0, \infty)$, with J_s fixed. The annealed state is the normalized full trace at $\beta = \mu = 0$: sector weights are $b_m = 2^{-N} \binom{N}{m}$. Write $h = H_m$, $d_m = \binom{N}{m}$, $\mathrm{tr}_m = \mathrm{Tr}\,/d_m$. For $m > 0$ set

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \qquad K = 1_N \otimes H_{m-1}, \qquad Q = 1 - VV^\dagger, \tag{2}$$

$$a = V^\dagger K V = (1 - p/m)h, \qquad R = QKV, \qquad D = QKQ, \tag{3}$$

$$G = R^\dagger R, \qquad M(s) = R^\dagger e^{iDs} R, \qquad \Lambda(s) = R^\dagger e^{iKs} R. \tag{4}$$

All sectors use the same J. Let $\mathsf{P}_h X = \sum_e P_e X P_e$, including whole degenerate energy blocks. The actual dynamic field is

$$\mathsf{F}_m(s) = (1 - \mathsf{P}_h)\big[m e^{-ihs/2} M(s) e^{-ihs/2} \big]. \tag{5}$$

Inactive $m < p$ sectors, including $m = 0$, are zero. Define convolution $A * B(s) = \int_0^s A(s-t) B(t)\, dt$, $U(s) = e^{ias}$ and $J_n = \Lambda * (U * \Lambda)^{*n}$, with $J_0 = \Lambda$. Replacing M by $J_0 + J_1 + J_2$ defines $\mathsf{F}^{[2]}$. We prove, uniformly for $|s|, |u| \le S$ and all real T,

$$\|\mathsf{F}(s) - \mathsf{F}^{[2]}(s)\|_{2,\mathrm{ann}} = O_S(p^{-2}), \tag{6}$$

$$|C_N(s, u; T) - C_N^{[2]}(s, u; T)| = O_S(p^{-1}), \tag{7}$$

where $C_N = \sum_m b_m \mathbb{E}\, \mathrm{tr}_m\, \mathsf{F}_m(s)^\dagger e^{iTh} \mathsf{F}_m(u) e^{-iTh}$ and $C_N^{[2]}$ uses the proxy fields.

*AI research assistant (OpenAI).

2 Exact finite iteration, not a time Taylor truncation

Removing projected propagation by auxiliary kernels is an established Mori method [1, 2]. Here finite matrices allow a direct proof. In $P \oplus Q$, identify P with the source space using V:

$$K = \begin{pmatrix} a & R^\dagger \\ R & D|_Q \end{pmatrix}. \tag{8}$$

The Q-block Duhamel equations give two transitions and hence $i^2 = -1$:

$$\Lambda = M - M * U * \Lambda, \qquad M = \Lambda + M * U * \Lambda. \tag{9}$$

For example, eliminating the P-block of $e^{iKs}R$ yields a convolution of e^{iDs}, R, U, $R^\dagger$, and the full Q-block. Multiplying by $R^\dagger$ and R gives (9), preserving the matrix order. Associativity gives, for any fixed integer $j \geq 0$,

$$M = \sum_{n=0}^{j} J_n + \mathcal{E}_j, \qquad \mathcal{E}_j = M * (U * \Lambda)^{*(j+1)}. \tag{10}$$

This is a finite identity with an exact remainder. No convergence of an infinite Neumann or Taylor series is assumed.

Normalize every rectangular Schatten trace by the *same* source dimension d_m. An integrand in $\mathcal{E}_j$ contains $2j+4$ factors R or $R^\dagger$, separated by unitary propagators. Schatten Hölder with exponent $4j+8$ bounds its Hilbert–Schmidt norm by $(\mathrm{tr}_m\, G^{2j+4})^{1/2}$. The $2j+2$ integrations have simplex volume $|s|^{2j+2}/(2j+2)!$. Therefore, samplewise,

$$\|\mathcal{E}_j(s)\|_{2,m} \leq \frac{|s|^{2j+2}}{(2j+2)!}(\mathrm{tr}_m\, G^{2j+4})^{1/2}, \tag{11}$$

$$\|J_n(s)\|_{2,m} \leq \frac{|s|^{2n}}{(2n)!}(\mathrm{tr}_m\, G^{2n+2})^{1/2}, \tag{12}$$

$$\|M(s)\|_{2,m} \leq (\mathrm{tr}_m\, G^2)^{1/2}. \tag{13}$$

For negative times use adjoints: reversing the simplex order gives $J_n(-s) = J_n(s)^\dagger$. Equivalently the oriented convolutions contain an even number of integrations. Thus all bounds apply to both time signs.

An elementary sign check is $K = \sigma_x$, $V = (1,0)^t$: $a = 0$, $R = (0,1)^t$, $M = 1$. The first terms are

$$J_0 = \cos s, \qquad J_1 = \frac{s \sin s}{2}, \qquad J_2 = \frac{s \sin s - s^2 \cos s}{8}. \tag{14}$$

In particular $M - \Lambda = s^2 G^2/2 + O(s^3)$ for general matrices. The finite-return expressions themselves retain all time orders.

3 Force square functions and uniform high moments

Let $F_i = [H, c_i]$, $\mathcal{A} = \sum_i F_i^\dagger F_i$. The exact dilation identity already used in the project is

$$G = \mathcal{A}_m/m - (p/m)^2 h^2, \qquad 0 \le G \le \mathcal{A}_m/m. \tag{15}$$

Here a direct Gaussian bound replaces any need to count high-order marked chord diagrams. For $T_{IJ} = c_I^\dagger c_J$, $[T_{IJ}, c_i]$ vanishes unless $i \in I$; otherwise it is $\pm c_{I\setminus i}^\dagger c_J$. The covariance maps of F_i are diagonal in occupation states. At charge m and occupation $\eta = n_i$, their eigenvalues divided by w_N are

$$f_{\mathrm{in}}(m,0) = \binom{m}{p}\binom{N-m+p-1}{p-1}, \tag{16}$$

$$f_{\mathrm{in}}(m,1) = \binom{m-1}{p}\binom{N-m+p}{p-1} + \binom{m-1}{p-1}\binom{N-m+p-1}{p-1}, \tag{17}$$

$$f_{\mathrm{out}}(m,\eta) = \binom{m-\eta}{p-1}\binom{N-m+p-1}{p}. \tag{18}$$

Invalid binomials are zero. Count the annihilated J first for f_{in}, separating whether it includes i; count the occupied $I \setminus i$ first for f_{out}. Overlaps $I \cap J$ are retained. Both counts are at most $(N+p)^{2p-1}/[4^{p-1}p!(p-1)!]$: bound binomials by powers, pair $p-1$ powers with $xy \le (x+y)^2/4$, and bound the remaining power by $N+p$. Hence

$$\mathbb{E}F_i^\dagger F_i,\ \mathbb{E}F_i F_i^\dagger \ \le v_N 1, \quad v_N = \frac{4J_s^2}{\lambda_N}\frac{p}{N}(1+p/N)^{2p-1} \le \frac{4J_s^2}{\lambda_N}\frac{p}{N}e^{2\lambda_N}. \tag{19}$$

For clarity, the Gaussian matrix moment step is also explicit. If $X = \sum_\alpha g_\alpha B_\alpha = X^\dagger$ and $\sum B_\alpha^2 \le v1$, Gaussian integration by parts gives

$$\mathbb{E}\,\mathrm{Tr}\,X^{2k} = \sum_{\ell=0}^{2k-2} \mathbb{E}\sum_\alpha \mathrm{Tr}\,B_\alpha X^\ell B_\alpha X^{2k-2-\ell} \le (2k-1)v\mathbb{E}\,\mathrm{Tr}\,X^{2k-2}. \tag{20}$$

To see the inequality, diagonalize X, take absolute values and apply weighted arithmetic–geometric mean to $|x_b|^\ell |x_a|^{2k-2-\ell}$. Symmetry of $|(B_\alpha)_{ab}|^2$ gives $\mathrm{Tr}(\sum B_\alpha^2)|X|^{2k-2}$. Apply the recurrence to the Hermitian dilation of the rectangular $F_i : \mathcal{H}_m \to \mathcal{H}_{m-1}$, stopping at its second moment. The factor two from the dilation cancels, so

$$\mathbb{E}\,\mathrm{tr}_m(F_i^\dagger F_i)^k \le (2k-1)!!v_N^k, \qquad \mathbb{E}\,\mathrm{tr}_m\,\mathcal{A}_m^k \le (2k-1)!!(Nv_N)^k. \tag{21}$$

The last step is Minkowski in the annealed Schatten k norm; independence between the different F_i is unnecessary.

4 Charge summation and the collective covariance error

Equations (19)–(21) hold uniformly in charge. For $k \geq 2$, eigenvalue monotonicity of $0 \leq G \leq \mathcal{A}_m/m$ gives $\operatorname{tr}_m G^k \leq m^{-k} \operatorname{tr}_m \mathcal{A}_m^k$; this does not assert that the power map is operator monotone. Eventually $p \leq N/4$. Split the binomial sum at $m = N/4$ and use $\Pr(m < N/4) \leq e^{-N/8}$. Since active sectors have $m \geq p$,

$$B_{k,N} := \sum_m b_m \mathbb{E} m^2 \operatorname{tr}_m G^k \tag{22}$$

$$\leq (2k-1)!!(N v_N)^k \left[(N/4)^{2-k} + p^{2-k} e^{-N/8}\right] = O_k(p^{4-k}). \tag{23}$$

There is no central-window limit exchange in this argument. Only $k = 2, 4, 6, 8$ are needed below. Exterior centering is unitary on both sides and $1 - \mathsf{P}_h$ is an orthogonal Hilbert–Schmidt projection. Thus, denoting the centered, pinched mJ_n by X_n,

$$\|\mathsf{F}(s)\|_{2,\mathrm{ann}} \leq \sqrt{B_{2,N}} = O(p), \tag{24}$$

$$\|X_n(s)\|_{2,\mathrm{ann}} \leq \frac{S^{2n}}{(2n)!}\sqrt{B_{2n+2,N}} = O_S(p^{1-n}), \tag{25}$$

$$\|\mathsf{F}(s) - \mathsf{F}^{[j]}(s)\|_{2,\mathrm{ann}} \leq \epsilon_{j,N} := \frac{S^{2j+2}}{(2j+2)!}\sqrt{B_{2j+4,N}} = O_S(p^{-j}). \tag{26}$$

Here $\mathsf{F}^{[j]} = \sum_{n=0}^{j} X_n$. The exact finite-$N$ energy pinching is retained in every term.

Conjugation by e^{iTh} is an isometry, including for sample-dependent fields. Cauchy–Schwarz on the direct sum and Gaussian probability space gives

$$|C_N - C_N^{[j]}| \leq 2\sqrt{B_{2,N}}\,\epsilon_{j,N} + \epsilon_{j,N}^2. \tag{27}$$

At $j = 1$, the field error is $O(p^{-1})$, but this only proves an $O(1)$ covariance error using the available field bound. At $j = 2$, (27) is $O(p^{-1})$, proving (6)–(7). An $O(1)$ bound on the true finite-time field was not presumed.

There is also a six-product form of the remaining calculation. Define

$$\langle X, Y\rangle_T = \sum_m b_m \mathbb{E} \operatorname{tr}_m X_m^\dagger e^{iTh} Y_m e^{-iTh}.$$

Then

$$C_N(s,u;T) = \sum_{\substack{n,l\geq 0\\ n+l\leq 2}} \langle X_n(s), X_l(u)\rangle_T + O_S(p^{-1}). \tag{28}$$

The omitted pairs $(1,2),(2,1),(2,2)$ are bounded by $O(p^{-1}), O(p^{-1}), O(p^{-2})$. For numerical positivity, the full nine-product $C_N^{[2]}$ remains preferable. All estimates fix S and $\lambda_- > 0$; they do not cover growing memory times or a simultaneous $\lambda \to 0$ limit.

5 Ordinary propagation and a stable spectral formula

Let $F(s) = V^\dagger e^{iKs}V$ (this F is a compressed propagator). Expanding $R = KV - Va$ gives the exact identity

$$\Lambda(s) = -F''(s) + i\{a, F'(s)\} + aF(s)a. \tag{29}$$

The sign of the last term follows directly from this expansion; it should not be copied from a different Mori projection convention. Keep $a = (1 - p/m)h$ exactly.

Choose eigenbases $h|a\rangle = E_a|a\rangle$ and $H_{m-1}|\alpha\rangle = \varepsilon_\alpha|\alpha\rangle$. Put $a_a = (1 - p/m)E_a$, $c^i_{\alpha a} = \langle\alpha|c_i|a\rangle$, and

$$W^\alpha_{ab} = \frac{1}{m}\sum_i (c^i_{\alpha a})^* c^i_{\alpha b}(\varepsilon_\alpha - a_a)(\varepsilon_\alpha - a_b). \tag{30}$$

Then $\Lambda_{ab}(s) = \sum_\alpha W^\alpha_{ab} e^{i\varepsilon_\alpha s}$. For $r \geq 0$, define the time-simplex integral

$$\mathcal{I}_r(s; \xi_0, \dots, \xi_r) = \int_{\substack{t_j \geq 0 \\ \sum t_j = s}} e^{i\sum_j \xi_j t_j}\, dt_0 \cdots dt_{r-1} = i^{-r}[e^{is\,\cdot}]^{[r]}(\xi_0, \dots, \xi_r), \tag{31}$$

initially for $s \geq 0$ and then continued to real s. The first two returns are the finite spectral paths

$$(J_1)_{ab} = \sum_{\alpha,\beta,c} W^\alpha_{ac} W^\beta_{cb} \mathcal{I}_2(s; \varepsilon_\alpha, a_c, \varepsilon_\beta), \tag{32}$$

$$(J_2)_{ab} = \sum_{\alpha,\beta,\gamma,c,d} W^\alpha_{ac} W^\beta_{cd} W^\gamma_{db} \mathcal{I}_4(s; \varepsilon_\alpha, a_c, \varepsilon_\beta, a_d, \varepsilon_\gamma). \tag{33}$$

Coincident energies use confluent divided differences. One stable evaluation is the top-right entry of e^{sB}, where B has diagonal $i\xi_j$ and unit superdiagonal. It automatically gives $s^r e^{i\xi s}/r!$ when all nodes coincide.

Set $L = J_0 + J_1 + J_2$. For one sector and one realization the proxy covariance is

$$\frac{m^2}{d_m} \sum_{E_a \neq E_b} e^{iT(E_a - E_b) + i(s-u)(E_a + E_b)/2} \overline{L_{ab}(s)} L_{ab}(u). \tag{34}$$

Summing with b_m and averaging gives $C^{[2]}_N$. Equal-energy blocks, not merely diagonal entries, are omitted. The phase in (34) keeps both the center time and the two memory times. No independent-copy replacement of J, annealed spectral factorization, or replacement of the exact pinching has been made.

6 Checks and the next calculation

The new driver `memory_return_expansion.py` in `code/` stores `finite-checks.json` in the stage results directory, without overwriting existing evidence. It checks:

- Five complete CAR systems through $N = 5$, including $p = 3$: 864 force-covariance diagonal entries agree exactly as integers; off-diagonal covariance entries vanish. Another 118,132 rational cases verify the combinatorial upper bound for $2 \le N \le 70$.
- Four common-coupling systems $(N, p, m) = (3, 1, 2)$, $(4, 2, 3)$, $(5, 2, 3)$, $(6, 2, 3)$ at $s = -0.8, 0.35, 1.1, 2$: all 48 finite-return remainder bounds hold. The derivative identity, negative-time adjoints and covariance phases are independently checked.
- A block-triangular exponential computes the return convolutions; a separate spectral-path sum checks J_0, J_1, J_2 at $(3, 1, 2)$. Its largest absolute-scaled matrix discrepancy is 9.39×10^{-16}. Five scalar cases check the explicit trigonometric formulas, including the confluent zero-energy return nodes.

The Hamiltonians in the finite propagation tests are seeded Gaussian integer Hermitian coefficient matrices, divided by their operator norms. Their seed, exact coefficient hash and scale are saved. They test identities, not ensemble asymptotics. Arithmetic is `complex128`; 2×10^{-12} identity tolerances and 2×10^{-13} bound allowances are empirical roundoff tolerances.

At $s = 2$, the observed remainder/bound ratios are:

N	p, m	zero returns	one return	two returns
3	1,2	0.852425	0.877750	0.899381
4	2,3	0.783725	0.838456	0.871689
5	2,3	0.913060	0.934683	0.947976
6	2,3	0.899579	0.921893	0.936414

The $(3, 1, 2)$ dynamic pinched covariance is numerically zero; the other three cases supply nonzero checks. For $(4, 2, 3)$ at $(s, u, T) = (-0.8, 1.1, 0.67)$, the two-return covariance error is 1.71146×10^{-5}, below the direct Cauchy–Schwarz bound 2.47819×10^{-5}. These finite errors are retained.

Next step. Evaluate the six common-coupling, pinched products in (28), establish finite-s tightness and identify their limit. This is a concrete route from the microscopic bilocal to the Susskind-line bulk meson problem. The present reduction does not yet determine that limit or a bulk mass/space dictionary. The earlier three Taylor-coefficient spectra remain unchanged.

References

[1] A. Montoya-Castillo and D. R. Reichman, *Approximate but Accurate Quantum Dynamics from the Mori Formalism: I. Nonequilibrium Dynamics*, arXiv:1603.01903, Sec. III, Eqs. (11)–(26). Auxiliary-kernel removal and derivative rewritings are established methods; that paper credits Shi–Geva.

[2] C. Widder, J. Zimmer and T. Schilling, *On the generalized Langevin equation and the Mori projection operator technique*, arXiv:2503.20457, Secs. 2–3. General context for projected evolution and Volterra equations; no unbounded-operator theorem is required here.